



Law and Economics in the Age of Computers

What are the implications of computational complexity for law and economics?

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Yale, Columbia and NBER

The Origins of Law

What Economics?

New school economics builds on contributions of a number of Nobel prize winners

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Overview of Course Content

Some Final Observations on the Use of Data (if there is time).

References

The Origins of Law

- Small groups routinely manage shared resources (fisheries, pastures, irrigation, forests) **without formal law** — avoiding the “tragedy of the commons.”
- Neither state control nor private property is required: **communities self-govern**.

Success rests on informal institutions that share common design principles:

- Clearly defined **boundaries** — who may use the resource, and its limits.
- Rules **fitted to local conditions**, with benefits matched to costs.
- Those affected **participate** in making and changing the rules.
- **Monitoring** by the users themselves or agents accountable to them.
- **Graduated sanctions** — penalties escalate with repeated violations.
- Cheap, local **conflict-resolution** mechanisms.
- Outside authorities **recognize the group's right** to organize.
- Cooperation is sustained by **repeated interaction, reputation, trust, and reciprocity**.
- See also Elinor Ostrom, *Governing the Commons* (1990). Nobel Prize 2009.

Formal Law Evolved in Large Societies.

- Ostrom's informal institutions work because members **know one another** and interact **repeatedly** — reputation and trust do the enforcing.
- As societies grow, exchange increasingly occurs **among strangers**, over distance and time, where informal sanctions and reputation break down.
- Scale brings **larger stakes and more complex trade**: land, harvests, loans, inheritance, and hired labor must be divided among many competing claimants.
- **Formal law** — written rules enforced by the state — supplies the impersonal, predictable enforcement that strangers need to transact.
- The central point: these early legal codes **evolved to allocate economic resources** — defining property, contracts, prices, and liability — not merely to punish wrongdoing.

(Greif et al. 1994; Ellickson 1986)

Mesopotamia: An Early Large Society

- Mesopotamia emerged around 3000 BC in the fertile crescent between the Tigris and Euphrates rivers - much of modern day Iraq. It was a society based on agriculture, trade, and markets. It required formal legal systems to manage disputes, enforce contracts, and protect property rights.
- Under Hammurabi there were about a million to a million and a half inhabitants, with a GDP that was about 30% - 40% higher than Europe at the time.

Hammurabi's Code



- Hammurabi's Code is one of the earliest legal codes, dating back to around 1750 BC in ancient Mesopotamia. It is a surprisingly comprehensive set of laws that covers various aspects of society, including family, property, immorality, theft, burglary, wage and price regulation, and even professional conduct.
- See Vincent (1904) for comprehensive discussion of the code.

Implications of Hammurabi's Code for Today

- The code is famously known for its principle of “an eye for an eye,” which emphasizes the idea of retributive justice.
- It also includes provisions for social hierarchy, with different punishments based on the social status of the offender and the victim.
- What is interesting is that many of the main issues have not changed in 3,778 years!
- What has changed are:
 - New technologies that requires new notions of property rights.
 - Better economic theory and evidence to guide the design of better rules (the goal of this course).

The Declaration of Independence (alt) states We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness.

These rights are protected by the rule of law:

- 5th Amendment (ratified 1791) — the due process clause:

“No person shall...be deprived of life, liberty, or property, without due process of law; nor shall private property be taken for public use, without just compensation.”

- 14th Amendment, Section 1 (ratified 1868):

“All persons born or naturalized in the United States, and subject to the jurisdiction thereof, are citizens of the United States and of the State wherein they reside. No State shall make or enforce any law which shall abridge the privileges or immunities of citizens of the United States; nor shall any State deprive any person of life, liberty, or property, without due process of law; nor deny to any person within its jurisdiction the equal protection of the laws.”

Keep in mind that there are two faces of the law.

- Control rights can be used for different purposes:
 - Legal rule can allocate power to the powerless to ensure that their rights are protected, and that they can bring suits against the powerful.
 - However, legal rules can also support what Acemoglu et al. (2001) call *extractive institutions* that gives rights to powerful individuals. These rights in turn allow them to extract resources from the weak (American slavery is an obvious example).

- All large societies have a complex of set of rules, law, along with a system of adjudication that have evolved over hundreds of years.
- For modern economics these laws are so complex no single person understands *all* laws.
- Existing laws are certainly not perfect — they are constantly evolving to respond to new economic and social challenges.
- The question then is how should be be changed, and what are the consequences of changes to a law, as opposed to the *hoped for* consequences.

What Economics?

What is Economics?

- Many scholars point to Adam Smith's *Wealth of Nations* (1776) as the beginning of modern economics.
- The key feature is to create a simple, mechanistic model of the economy that allows one to understand how the economy works, and to predict the consequences of changes in the environment.
- It is a mistake to think that science is about finding the “true” model of the world — The world is simply too complex.
- The goal of science is to build simple models that humans can use to make better decisions (for example weather prediction, earthquake prediction or determining the impact of a change in the law).

- Law and economics movement started with the founding of the *Journal of Law and Economics* by Aaron Director and Ronald Coase at the University of Chicago.
- The textbook Posner (2014) was very influential in bringing price theory to the law school.

The original work in law and economics was very much *old school* economics (aka “Chicago School of Law and Economics”)

- The economy is viewed from the perspective of *price theory* - with free market prices efficiently allocate resources.
- Posner argued that legal rules can be explained as (imperfect) wealth maximizing solutions.
- The main barriers to the efficient operation of the economy are government overreach and monopolies, such as unions and monopolized industries.

Old school is necessary but not sufficient.

- Price theory is a very useful first order approach to understanding both the economy and the law.
- In addition to Posner's book, both Cooter and Ulen (2016) and Shavell (2004) also provide excellent treatments of law and economics.
- These books use the intuition of price theory, combined with game theory (see Baird and Gertner (1994)) and transactions cost economics (Williamson (1985)) to provide a framework for the study of legal rules.

New school economics builds on contributions of a number of
Nobel prize winners

New School Economics I: Failures of Price Theory due to Imperfect Information

Year	Laureates	Contribution Summary
2001	George A. Akerlof, A. Michael Spence, Joseph E. Stiglitz	For fundamental analyses of markets with asymmetric information (when buyers or sellers have unequal information).
2010	Peter A. Diamond, Dale T. Mortensen, Christopher A. Pissarides	For analyzing markets with search frictions (helping explain why high unemployment can coexist with job vacancies).
2016	Oliver Hart, Bengt Holmström	For foundational contributions to Contract Theory (analyzing executive pay, performance contracts, and privatization).
2007	Leonid Hurwicz, Eric S. Maskin, Roger B. Myerson	For establishing Mechanism Design Theory (how institutions achieve desirable economic outcomes when information is private).

Year	Laureates	Contribution Summary
2002	Daniel Kahneman, Vernon L. Smith	For integrating psychological research into economic decision-making under uncertainty (Kahneman) and establishing empirical laboratory experiments (Smith).
2017	Richard Thaler	For incorporating psychologically realistic assumptions into economic decision-making (Behavioral Economics).

New school Economics III: Analysis of Big Data (Causal and Inductive)

Year	Laureates	Contribution Summary
2021	David Card, Joshua Angrist, Guido Imbens	For empirical contributions to labor economics (Card) and methodological breakthroughs in analyzing causal relationships using natural experiments (Angrist/Imbens).
2024	John J. Hopfield & Geoffrey E. Hinton	Foundational Neural Network Architectures: Awarded for foundational discoveries that enable machine learning with artificial neural networks. Hopfield created an associative memory (Hopfield Network) that can store and reconstruct patterns. Hinton built on this to invent the Boltzmann machine, paving the way for deep learning architectures that underpin modern generative AI models.

For the purposes of this course, New School Economics is divided into three groups (see MacLeod (2022) for a review of the theory):

- Price Theory and Game Theory (1940-1980) - the foundations of modern economics.
- Information Economics, Contract Theory and Causal Analysis (1980-2010)
- Bounded Rationality, Behavioral Economics and Big Data (2010-now)

Law and Economics in the Age of Computers

- Quickly review price theory, information theory and bounded rationality/behavioral economics.
- Show how contract, tort, crime and property laws are designed to improve economic performance.
- Discuss how one can use economics to evaluate the consequence of changes in the law.
- Finally, discuss how AI's impact on reducing bounded rationality leads to the need for new laws and regulations.

AI researchers have invented the term *The Alignment Problem* (Christian (2020)) to refer to these problems.

- The alignment problem is often framed in terms of making the AI systems safe. Yet, the fundamental problem is not the AI system, but the humans that use the system. For example, a gun only becomes dangerous when it is used by a human. We have gun laws to control the way humans are able to access and use dangerous weapons such as guns.
- The last homework project of this course asks you to consider the alignment problem in the context of governance structures of AI firms. We will discuss your project reports with Professor John Morley, who works with Anthropic on governance issues.

Overview of Course Content

- Economics -- Review price theory, information theory and bounded rationality research in economics. behaviors.
- Contracts -- the foundation of exchange.
- Tort -- *private law* enforcement of *civil* behavior.
- Crime -- *public enforcement* of *civil* behavior.
- Property -- What is property? When is a firm property that can be bought or sold, or a legal person? Who is responsible for AI technology? Are AI agents legal persons? If not who owns them? See Hohfeld's framework for understanding property rights ((?)).

Deliverables

- Weekly reading of slides and one paper/case per week (2 points per week, but by noon of the day of the lecture - 0.5 bonus points for comments submitted the previous day before midnight - points for the first week do not count towards the total).
- A midterm just before the midterm break (10 points) and final exam (20 points) based upon the questions at the end of each lecture's slides.
- Yale curriculum committee requires a writing assignment. Students may choose to write a 10 page paper over the midterm break rather than take the midterm exam (10 points). If one chooses both, the your mark will be the maximum of the two.
- Four group assignments on contracts, tort, crime and property (5 points each). Students are required to meet at least twice over zoom (zoom summary is submitted) and to deliver a 10 page report (references included). The groups are free to use AI to help with the report. The goal is to build group skills that are essential for the modern workplace.

- Some notes on AI usage are posting on the teaching page or my webside (www.wbmacleod.net).
- Note that you can always ask Claude of other system on how to use effectively.
- Are there questions?

Sone Final Observations on the Use of Data (if there is time).

- The law has always been an empirical discipline:
 - Plaintiffs bring cases to the courts.
 - The courts listen to evidence and adjudicate based upon this evidence.
- Given legal norms are evolved over centuries, they can be difficult to change, and have a language and modes of thinking that different from modern science.

- With the advent of modern computers and high quality data set, empirical methods in economics have evolved very quickly, particularly since 1980.
- The consequence of better data and methods is modern economics is much less ideological.
- Modern economics has a greater focus on estimating the causal effect of changes to the environment.

Evidence is of increasing importance

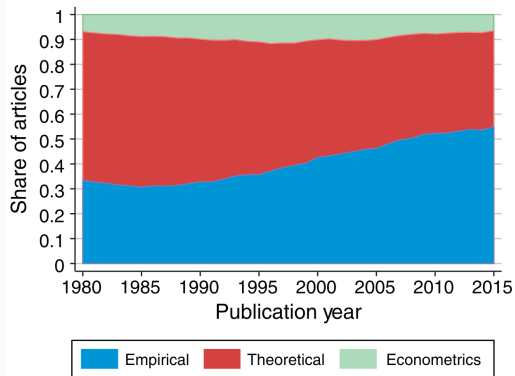


FIGURE 6. WEIGHTED PUBLICATIONS BY STYLE

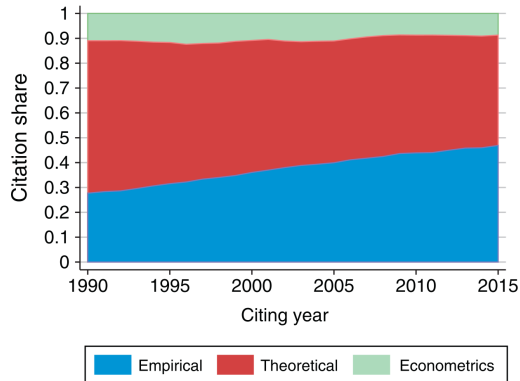
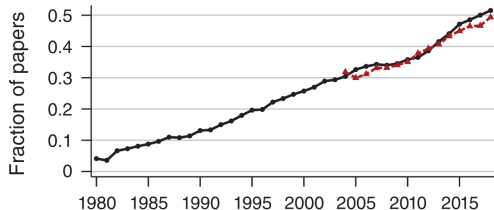


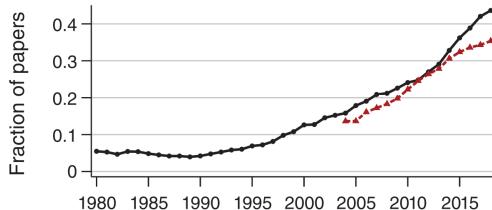
FIGURE 7. WEIGHTED CITATIONS BY STYLE

More types of evidence are now available to evaluate the impact of legal rules and institutions.

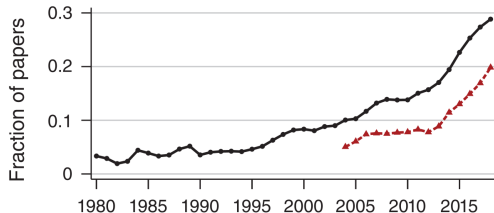
Panel A. Identification



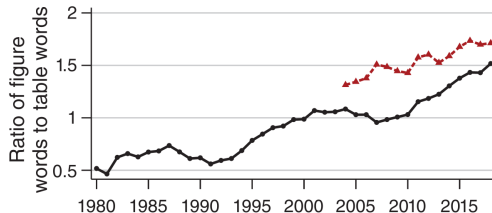
Panel B. All experimental and quasi-experimental methods



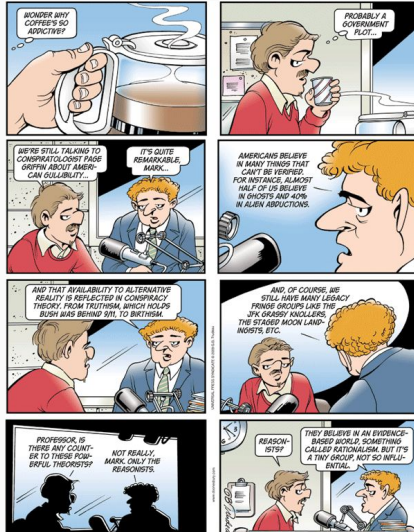
Panel C. Administrative data



Panel D. Graphical revolution



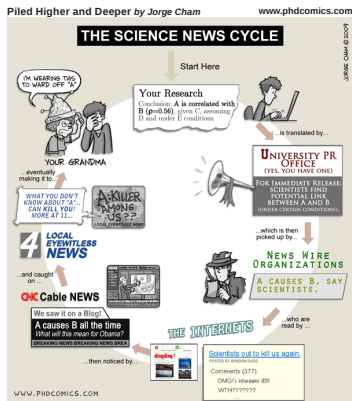
How not all humans use evidence based decision making - I.



How not all humans use evidence based decision making - II.

Piled Higher and Deeper

http://www.phdcomics.com/comics/archive_prin...



- What is the alignment problem in AI? How is it related to the design of legal rules and institutions?
- What is the difference between informal and formal law? How do they relate to the size of a society?
- What is the difference between extractive and inclusive institutions? How do they relate to the design of legal rules and the allocation of decision rights?

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